Pointless Coherence: Order Without a Centre

Patrick S. Mahon

October 2026 (draft)

The question. When do local views, each correct as far as it goes, add up to one world? And when they do not, what are they still telling us?

Part I gives the mathematical and philosophical core. Part II asks where the question is lived: in ecosystems with no species on top, and in societies with no one in charge. Part II is a lens, not an application, and Section 16 says plainly how far the mathematics reaches. Status labels: [Lean] machine-checked, [hand] proved by hand, otherwise conjecture.

Part I. The essence

1 The stance

Answers come first. Questions, and patterns of answers to them that are consistent, are taken as primitive. A *world* is something answers have to *earn* by fitting together. It may be an actual point, or an ideal point at infinity. Nothing is assumed to stand behind the answers except the bare space of possibilities.

The patterns studied here are *complete* (every question that can be asked gets a yes or a no) and *coherent* (answers to disjoint questions add up). In ordinary logic, complete and coherent always means there is a world: the completeness theorem [18]. That relies on every "and" being askable. The theory studies what happens when it is not, and so it studies one gap:

complete and coherent, yet about no world.

2 The picture

Penrose's staircase [38]. Each flight of stairs is a *query*: a view of two landings that says "this one is higher". (Physicists call such a view a *measurement*, and the pattern of overlapping views a *measurement scenario*.) Every flight goes up, and the flights close into a loop. Each view is fine on its own, but no building satisfies all of them. The *questions you can ask* are the questions about one flight at a time, and anything built from those by "not" and by joining questions that never both say yes. Joint questions across flights are not automatically askable. Formally this collection is the *carrier*, in the literature a σ-class or concrete logic [35].

Terminology. The puzzles use plain words; the status-bearing statements use the technical ones.

plain	technical
landing, height	variable, value
flight, query, view	pair context (a measurement context on two variables)
the questions you can ask	the carrier: the Dynkin system generated by the pair events
coherent answer pattern	a state on the carrier (finitely additive, normalised, non-negative)
a building; a world	a point of Ω; a point or ideal point (ultrafilter), or for probabilities a charge on all subsets of Ω
a mixture of real buildings	a convex combination of point masses: a joint distribution whose pair marginals are the given laws (the laws lie in the marginal polytope; in contextuality, a global section)
a sum of local facts	an element of the linear span of the pair indicators
a twist	pair laws consistent on overlaps with no global extension (a gluing obstruction; contextuality)
visible / hidden twist	the forced functional is negative on some carrier set / is a state on the carrier but not a mixture of points
pointless	a two-valued state whose true sets lack the finite intersection property, i.e. no ultrafilter on Ω induces it (not pointless topology; on infinite Ω not Pták's "free"; weaker than De Simone–Navara–Pták non-extendability)

3 What is known

regime	coherent answers earn a world?	status
finitely many possible heights	yes, an actual one	[hand]
infinitely many, answers add over the whole infinity	yes, an actual one	[hand; Lean on rings]
infinitely many, finite adding, no loop	yes, an ideal one	[hand]
infinitely many, finite adding, a loop	**no**: pointless answers exist	[Lean + hand]

Theorem. *Coherent answers can be about no world exactly when the views form a loop, there are infinitely many possibilities, and answers add up only finitely.*

What separates the finite from the infinite case. A yes-or-no pattern is about an (ideal) world exactly when its true answers have the finite intersection property. A pointless pattern says yes to finitely many questions that are pairwise compatible yet jointly impossible: Specker's seer, realised at infinity. Finitely many heights cannot do this, and countable adding restores the property [hand].

In Aristotle's terms [7], finite adding is the *potential* infinite. Demanding the actual, completed infinity forces a world back into existence. "Ideal" and "pointless" match the standard notions of coherent and non-coherent finitely additive states [9]. For ordinary probabilities and finitely many outcomes, the tree/loop line is classical [45, 8]; so is the method of carrying a loop's failure to any graph with a cycle (reduce to a triangle or an induced cycle, then fix the other coordinates) [44] [audit, October 2026]. Which patterns the askable questions catch is the exclusivity principle of contextuality theory (Section 7). What is offered as new lies on the infinite side.

4 Why: what the askable questions contribute

On the triangle (three landings, three flights) the contradiction could only be exposed by one question: *do the three heights go round the staircase's way?* That question can never be built, not even in infinitely many steps [Lean]. Two facts do the work.

- *Balance.* Pick two candidate heights for each landing. That gives the eight corners of a box. Every buildable question says yes equally often on the even and the odd corners.
- *The lift.* Raising the lowest landing above the highest never changes the answer to a buildable question, because the box joining the two configurations has its other six corners going the other way round.

So the coherence survives because the one question that would break it is unaskable. Beyond that one exclusion, whether answers are pointless depends only on whether the windows' local answers fit together. In every case checked, any local answers that agree where they overlap extend to every askable question [hand; Lean for the triangle cases]. The nearest precedent is Dravécký and Šipoš's example [13] and the work of Navara and Pták [35, 34]. There, answers that add over the whole infinity can be pointless on a smaller collection of questions, but they do not extend to all the askable ones.

5 What creates impossibility, and what does not

- *Renaming never does, for yes-or-no answers.* A window that only says "this landing is that one, renamed" can be contracted away [hand, via Katětov's lemma [23]]. With probabilities this fails: a loop of renamings that shifts every value, like "each landing is one more than the last", can be a twist, whether the values wrap round a finite clock or spread evenly over all numbers [computation; hand].
- *Motion never does.* Symmetry and walking the stairs only move worlds around. The twist lives in the loop, not in the action [hand].
- *"Beyond" can.* A window that draws one landing *beyond* the other carries a direction. A loop of such directions is a task that would have to finish before it started.
- *And beyonds compose.* A triangle whose directions never go round can still fail to close. See the next section.

6 The twist is a failure of composition

A *twist* is local agreement whose composition round a loop does not return to its start.

Two forward queries force the third. Call a query *forward* if it draws its second landing after the first, landing by landing, and each value of the second landing comes from only finitely many values of the first. Independent "beyond" draws, renamings and steps "$x_1 = x_0 + b$" are all forward. If the queries on $\{x_0, x_1\}$ and $\{x_1, x_2\}$ are forward, then any world fitting both is *unique*, and it fixes how x_0 relates to x_2: the *composite* of the two queries, an iterated sum of ultrafilters [19]. So the triangle has a world exactly when its third query *is* that composite [Lean]. Around a longer loop, the closing query must equal the composite of the others [hand].

What it explains. The twist is the *holonomy of composition*: the closing query disagreeing with what the rest compose to.

- *Independent beyonds:* the composite is again "x_0 first", so the twist is a directed cycle (for orientations with a forward path).
- *Renamings:* the composite is the composed renaming, so the twist is its monodromy (automatically trivial for yes-or-no answers).
- *Steps:* the composite of steps d_{01} and d_{12} is their ultrafilter sum. A triangle whose third query is an *independent* draw is therefore twisted for every non-principal ultrafilter, although nothing goes round in a circle [hand].

Prior art: the sum of ultrafilters is classical (Frolík), and the pair-level description of how two ultrafilters glue is Goldberg's [19]. The cases above look classical one by one; what appears new is the unification. Queries that are not forward are the open edge.

7 Probabilities and finite worlds

So far answers were yes or no. Let them be probabilities: each query reports how likely each pair of heights is, and the probabilities of the questions you can ask must add up. A *mixture of real buildings*, technically a convex combination of point masses, is what you get by choosing heights at random.

- *Loops without shortcuts are blind.* On a ring of four or more landings the questions you can ask are exactly the questions about one query, so any pattern that agrees where queries overlap is allowed, including impossible ones. Example, with two heights around a square: "different, different, same, different". Every query is fine, but going round flips the height three times [hand; certified example].
- *The triangle fills itself in, up to three heights.* With two or three possible heights, every state on the triangle's carrier *is* a convex combination of point masses [exact computation]. With four heights this fails: a sum of local facts taking the values 0, 1, 2 is an extreme non-negative witness that no askable question replaces, so a probabilistic twist can hide on a finite triangle [exact computation + Maharam's criterion [29]; corrected October 2026]. In fact the line is exact: with sizes sorted $r_1 \geq r_2 \geq r_3$, a triangle can hide a probabilistic twist exactly when $r_2 \geq 4$ and $r_3 \geq 3$, which is exactly when the margin polytope of three-way tables is not "compressed" [40, 41] [hand + exact finite computation, on owned theorems]. Reading that threshold as the point where every askable question goes blind is itself a short restatement of Sullivant's criterion plus Farkas separation [audit, October 2026]; what may be new is only that the questions built by "not" and "either one" already reach every 0/1 facet at the compressed sizes. Two triangles sharing a flight, four landings all joined, and a fan of three triangles behave the same way at two heights, while the five-ring and the "house" (a square with a roof) hide twists [evidence]. So in finite worlds shortcuts help but do not settle it: the number of heights matters too.
- *For yes-or-no answers, idealisation is what turns a shortcut into a twist.* With infinitely many heights and finite adding, the triangle carries pointless answers [Lean]. A box-by-box limit of convex combinations of point masses is again one [hand]. For yes-or-no answers, pointlessness needs infinity: with finitely many heights every coherent yes-or-no pattern is a point.
- *Renaming can twist, with probabilities.* For yes-or-no answers a loop of renamings never creates impossibility. With probabilities it can: a loop saying "each landing is one more than the last" is coherent on each query and impossible as a whole, whether the heights wrap round a clock or spread evenly over all numbers [computation; hand]. So the sharp divide is between *yes-or-no* and *probabilistic* answers, not between finite and infinite.

A third kind of twist. Renamings (gauge) and "beyond" (order) are not the only ways to twist. On a triangle whose heights take three, four and four values, let each pair of landings say only which combinations are *allowed*, nine combinations each, all equally likely. Every allowed pair extends to an allowed triple, so nothing is logically impossible. Yet no weighting of the triples reproduces the even spreads, and no askable question detects this [exact computation; certified with an integer program]. The twist is *relational*: it lives in many-to-many constraints that are fine enough. Composed pair by pair, such answers allow a noticeably larger set of closing laws than any world does (about a quarter more, by volume, in this example). The extra laws pass every askable question and almost every summary check, and show up only along special directions [computation].

The loop count. Five people sit in a ring, and each neighbouring pair reports "we differ", each person's answer split evenly. With two possible answers this is impossible: going round an odd ring, someone must agree with a neighbour, so the number of disagreements is 0, 2 or 4, never 5. Yet the reports pass every question you can ask. What exposes them is the *loop count*, $4 - \#\text{disagreements}$. It is never negative in any world, and the reports give it -1. The loop count is a *sum* of local facts, but no query sees it and it cannot be built from the questions you can ask.

- *Two ways to be made of local information.* A fact can *add up* from local facts (lie in their linear span), or be *built* from them by "not" and "either one". Built implies adds up, never conversely.
- *Lemma.* With finitely many possibilities, a state on the carrier is a convex combination of point masses exactly when it gives every non-negative sum of local facts a non-negative value. So a hidden twist is always exposed by a non-negative sum that cannot be built: either a yes-or-no question that adds up but cannot be built (on the four-ring the loop count is such a question), or a many-valued one (on the five-ring) [owned: Maharam [29], Thm 6.1; Horn and Tarski [17]; computation on eight small graphs]. Maharam's Example 2.1 is the triangle itself, in linear form.

This is contextuality, and it is known. The finite side of this section is owned by the quantum foundations literature.

- *The witnesses.* The loop counts are exactly the tight inequalities of the n-cycle scenario, $\sum_i \gamma_i \langle X_i X_{i+1} \rangle \leq n - 2$ with an odd number of signs $\gamma_i = -1$ [6]. The four-ring is the CHSH Bell test, and its twist "different, different, same, different" is the Popescu–Rohrlich box.
- *What the askable questions catch.* A pattern that respects the questions you can ask never gives more than total probability one to outcomes that pairwise contradict each other. That is the *exclusivity principle*, also called local orthogonality or consistent exclusivity [11, 15, 5]. On the triangle, the four-ring, the five-ring and the house, the two conditions cut out exactly the same patterns [exact linear programming, two answers per landing; not proved in general].
- *Rings hide, triangles do not.* A single n-cycle Popescu–Rohrlich box obeys the exclusivity principle for every $n \geq 4$ [12], while Specker's triangle violates it. That is the finite fact "rings of four or more hide twists". Choudhary and Barbosa show that several independent copies can expose what one copy hides, using finite Ramsey theory.

Where the new ground is. For yes-or-no answers, a triangle hides a twist only with infinitely many heights and finite adding [Lean]. For probabilities, hiding is finite and its

line is exact: it happens precisely at the sizes where the margin polytope is not compressed (above). Two things look new after the prior-art passes so far, though neither is claimed yet: that classical threshold, read as the place where the askable questions go blind; and the yes-or-no, infinite regime. The infinite proof also runs on Ramsey theory, for ultrafilters on triples of numbers. Whether "infinitely many values" and "many independent copies" are one lever is open, and it is an analogy, not yet a transfer.

Whether the infinite triangle's questions absorb such a probabilistic renaming twist depends on *which* questions are askable.

- If each landing's own value may be asked about, reduce modulo 2: the question "do the first and third landings have the same parity, and the second the other?" is askable, and the renaming loop gives it probability $-\frac{1}{2}$. The twist is visible [hand; exact computation].
- If only relations may be asked about, one difference at a time, the askable questions form Dravécký and Šipoš's class [13], and the twist hides there.

Section 8 takes this up: which questions are askable is a choice between relations and relata.

8 Asynchrony, and relations versus relata

Shared time forces a world. Let each view read a process with its own *clock*: an average over its own epoch, or its own long-run limit. If every view uses the same clock, their reports are the marginals of one joint law, so a world exists [hand, one line]. A failure to add up therefore needs *asynchrony*: views without a common clock or scale. Asynchrony is not enough. In the three-species cycle, stations reading rarity in decades all run off the cycle's lap counter, a hidden shared clock, and a world is forced [hand].

Clocks that cannot be shared. Relativity forbids a shared clock round a rotating loop. Synchronise clocks station by station round the equator and the loop does not close, by about 207 ns ($2\Omega A/c^2$; Einstein synchronisation has holonomy [30], and the Sagnac difference [39] is twice that). GPS resolves it by adopting a centre: the time of a non-rotating, Earth-centred frame. Read answers first, with each clock's absolute setting spread evenly, every pair agrees and no world fits. Only whether the holonomy is non-zero matters [hand]. The question becomes Gudder's [16, 13, 35, 34]: *does the expectation of a sum equal the sum of the expectations, when the observables cannot be measured together?*

Relations or relata. Whether that failure is visible depends on what may be asked.

- *Relations only*, one at a time (how clock 1 stands to clock 2): the answers are coherent and about no world. The twist is hidden. This is Dravécký and Šipoš's example [13].
- *Relata allowed* (what clock 1 itself reads, in units of the holonomy): ask whether clocks 1 and 3 have the same parity and clock 2 the other. The answers give this probability $-\frac{1}{2}$, so the twist is visible [hand; exact computation].

The same holds for rankings. Three pairwise majorities that run in a cycle (Condorcet) pass every question built from comparisons, even combined with questions about each candidate's own value: with three or more possible values, every such question is about a single pair, so no pair-by-pair record can be contradicted by it [proved; core Lean-checked]. Questions about each candidate's own score expose the cycle when the scores wrap round (a finite reduction carries the twist), and not when they run off to infinity [hand; Lean]. Asking about relata presupposes them, and when no world exists that presupposition is what turns a quiet incoherence into a visible one. This is a reading of two cases, not yet a theorem.

9 Three kinds of twist

Three small puzzles, one for each way a loop of pairwise views can fail to close.

- *Renamed clocks.* Clock 2 reads what clock 1 reads; clock 3 reads what clock 2 reads; clock 1 reads one unit more than clock 3. Each pair is fine; round the loop the labels do not line up. This is a *gauge* twist: a mismatch of renamings, measured by a holonomy.
- *Staircase.* Landing 1 is beyond landing 0, 2 beyond 1, 0 beyond 2. Each flight goes up; the loop cannot. This is an *order* twist: directions that close into a cycle.
- *Allowed pairs.* Each pair of variables says only which combinations are *possible*, nine of them, all equally likely. Every possible pair extends to a possible triple; yet no weighting of the triples reproduces the even spreads. This is a *relational* twist: many-to-many constraints whose frequencies do not fit together.

When can the questions you can ask see each kind?

	gauge	**order**	**relational**
hides on a triangle when	only relations may be asked about	answers are yes or no and values infinite	exactly when $r_2 \geq 4$, $r_3 \geq 3$ (sorted)
exposed by	a finite reduction (parity)	nothing askable (pointless)	nothing askable
status	hand; exact computation	Lean	certified computation
owned	relations-only case: [13]	triangle of types: model theory	threshold: [40, 41]

What does the detecting. Three further facts locate where incoherence can be seen.

- *Views, not relations or relata.* Questions about how things stand to each other, about what each is by itself, or any combination of the two, can be blind where a single question about what a pair shows *together* is not. For orderings this is a theorem: with three or more values, every question built from comparisons and individual values concerns one pair only [proved; core Lean-checked].
- *Composition.* Given two pair laws, the askable questions allow a set of closing laws that contains every realisable one. Below the $3 \times 4 \times 4$ threshold the two sets coincide; above it the askable set is larger, by a full-dimensional region that passes almost every summary check [computation].
- *The atom.* In every search at $4 \times 4 \times 4$, the most hidden incoherence takes the relational form: allowed pairs, evenly spread [computation; a candidate, not a theorem].

What is offered here is the classification and the reading in terms of askable questions; the separate ingredients are owned as cited.

10 The lineage

- *Specker* [43]: a seer's three boxes, any two of which can be opened consistently but not all three. That is the triangle, and the seed of the Kochen–Specker theorem [26]. The sheaf-theoretic account of contextuality [1] and the classification of cycle scenarios [25] own the finite, ordinary-probability side, and the exclusivity principle [11, 15] says which finite twists the askable questions catch.
- *Penrose* [38]: impossible figures as a property of loops. Here the loop is necessary, and the obstruction is the way steps compose.
- *Kant* [22]: the completed totality of conditions is not *given* but *set as a task*. An ideal point is that task, and a loop is where the task becomes circular.

Part II. The lived

11 The lizards

On the hills of central California, male side-blotched lizards come in three throat colours, each with its own strategy. Orange males hold large territories and take mates from blue males. Blue males guard their mates closely and so beat the yellow males. Yellow males sneak into orange territories and beat the oranges. Orange beats blue, blue beats yellow, yellow beats orange. The frequencies cycle over the years, and no colour wins [42].

That has the *shape* of Penrose's staircase: every flight goes up, the stairs close into a loop, and there is no top. But the lizards' loop is a loop of *relations* (who beats whom), measured as a payoff game [42]. The theory's twists live in loops of *views*: observers who each see two variables and report a pattern. The published lizard data contain no such views, so they cannot be tested for a twist; that would need data collected as pairwise views. Similar cycles are documented among toxin-producing strains of *E. coli*, where the cycle keeps all three strains alive when interactions are local [24], and among encrusting organisms on coral reefs [10]. Network models proposed that such cycles can sustain many more species than a ranking could [4]. A later replication found the effect much weaker [28], so the strong claim is contested.

Ordinary logic says a cycle of "beats" is a contradiction, because no ranking satisfies it. Ecology shows it can be an order. It is just not a *ranking*. That is the shape of the central finding of Part I: coherent answers with no single world behind them.

Others have read this cycle non-classically. Aerts, Sinervo and coauthors report a violation of Bell's inequality in the lizards' mating competition [2]. Their pairwise probabilities come from a fitted model rather than direct measurement, and a violated marginal law is what contextuality-by-default calls direct influence, not contextuality. No one, as far as we found, has run a test of whether a single joint description *exists* for such a cycle's pairwise data. That test is the honest version of the question, and it is open.

12 Ecology: order from local relations

Cycles of dominance. An ecosystem does not consult a global ranking. Each encounter is local: this lizard, that territory, this season. When local relations form a *tree*, they always add up to one global order. When they form a *loop*, they can be coherent with no global order at all.

Steps across scales. Holling's *panarchy* [20] describes nested adaptive cycles at different speeds: a leaf, a stand, a landscape. The finding that beyonds compose rhymes with this. The question is not only whether influences go round in a loop, but whether they *compose* across scales. It is a rhyme, not yet a theorem about ecosystems.

Partial observation. Ecological knowledge is assembled from plots, transects, traps and seasons, each a local view. Do my surveys connect in a loop? If not, local agreement is enough. If so, they can agree pairwise and still describe no single landscape.

13 Anarchism: order without a sovereign

The stance is already anarchist. The theory takes *answers first*. There is no presupposed ground and no sovereign "real world" that the answers report on. A single world, or

a single centre, is something local answers must *earn*. This mirrors an anarchist account of order: it is not handed down from a centre, it is made, or not made, from relations. Feyerabend's "epistemological anarchism" [14] makes the same refusal for science.

Hierarchies are trees. Local agreements arranged as a tree always add up to one global order. A chain of command is a tree, which is why hierarchies *look* coherent: structurally, they cannot fail to be. The cost is that coherence is bought by removing every relation that does not route through the centre.

Federations are loops. Kropotkin was a field naturalist before he was an anarchist. In *Mutual Aid* [27] he argued from animals, villages, guilds and medieval cities that cooperation built from local relations is as real a force in evolution as competition. Elinor Ostrom made the case empirically [36]. Swiss alpine pastures, Japanese village forests, Spanish irrigation huertas and Philippine irrigation societies governed shared resources for centuries with no state or market in charge. One of her design principles is *nested enterprises*: overlapping local centres of rule-making. Her *polycentric governance* [37] generalises it. In the theory's terms these are webs with loops. Pointless coherence is, structurally, *order without a sovereign*.

What a centre asks for. A web of relations can cohere with no underlying table of positions behind it. Demanding relata (a ranking, a common clock, each party's standing in itself) is what can turn such an order into a visible contradiction, and adopting a centre, as GPS does, is one way to restore a world. Both are choices about which questions to ask. This is a reading of the clock and ranking cases (Section 8), not a theorem about societies.

How commitments chain. Local agreements can be fine, even pairwise, yet fail to add up. That happens not because anyone contradicts anyone, but because the agreement between the first and last parties must match the composite of those in between.

14 Cases: testing the ideas against the world

The ideas are held to concrete cases before any generalisation, and the cases can push back.

- *Three species that never settle.* In a rock-paper-scissors cycle of competitors [31], long-run averages never converge; they circle a polygon forever [32, 33]. Three field stations, each watching one pair of species over its own span of time, can agree wherever they overlap and still imply that "no species is dominant" has probability -0.2: no long-run world fits them all [simulation and exact geometry]. Each long-run world exists only as an ideal limit, and there is a whole loop of them; the observer's choice of time span selects one.
- *One quantity in time windows.* Watching one quantity through overlapping time windows never produced the failure: windows that share the same variable share too much to disagree [simulation].
- *Three bodies.* Pairwise observers of the three-body problem could describe no world only along motions whose statistics never settle. How common those are is tied to a famous open question of Kolmogorov. In an isolated three-body universe each body's own view of its two relations forms the triangle, but Newtonian shared time forces a world [hand].
- *Rarity in the long run.* Let the field stations read how many decades down each species is. That number is unbounded only in the continuous, infinite-population

model; real populations go extinct [3]. The stations' reports all run off the cycle's lap counter, which acts as a shared clock, and for every station that reads a fixed fraction of each lap a world is forced [hand; the rest open].

- *Clocks round the Earth.* Relativity forbids a shared clock, and the synchronisation loop fails by 207 ns (Section 8). Visible if a clock's own reading may be asked about, hidden if only relations may [hand; exact computation].

The pattern so far, which is a description and not yet a law: no world needs views of separate variables, statistics that never settle, and observers who cannot share a clock or way of averaging. Shared time forced a world in every case that had it; relativity is the case where nature forbids it.

15 The four questions, lived

question	**in an ecosystem**	**in a commons or federation**
Is there a loop?	cycles of dominance, nutrient cycles, feedbacks	overlapping councils, reciprocal obligations
Is there a joint question nobody can ask?	"who is best?" when dominance cycles	"who is in charge?" in a polycentric order
Is an idealisation hiding a contradiction?	equilibrium, carrying capacity, "in the long run"	"the general will", "the market" as one agent
Circle, or composition?	do influences compose across scales?	do commitments chain consistently?

16 Honest distance

link	**status**
lizards, *E. coli*, reefs	*shape only*: a cycle of pairwise dominance has the staircase's shape, but it is a loop of relations, not of views; the published data cannot be tested
clocks round the Earth	*literal*: a measured synchronisation holonomy; visible or hidden depending on what is askable
relations versus relata	*reading* of two cases (clocks, rankings)
hierarchy as tree, federation as loop	*structural*: the forest/loop theorem applied to the shape of relations
answers-first as anarchist epistemology	*philosophical*: a shared refusal of a presupposed centre
panarchy and composition	*a rhyme*: suggestive, not yet a theorem

Part I works with idealised infinities. With finitely many possibilities its own theorem says every coherent pattern *is* a world. The lizard cycle is coherent as a *dynamics* rather than as a set of answers, and real commons are finite. So the lived reading needs a different bridge: a *probabilistic* or *dynamical* version, in which "locally consistent but globally unrealisable" is real for finite systems. Belief propagation on loopy networks is one such setting [46].

The finite version (Section 7) already changes the lived picture. Loops without shortcuts, such as a ring of four or more parties with no cross-links, can hold locally coherent arrangements that no whole supports. Shortcuts help but do not make a web tree-like: with four or more possible values, even a triangle can hold such an arrangement (corrected October 2026). For a commons or a food web, the structural question is not only whether its loops have shortcuts, but how fine-grained the local states are. Whether the composition law has a finite, probabilistic shadow is open, and it may be the bridge that matters most.

That finite bridge is real, and it is not ours: it is contextuality (Section 7). This is good news for practice. The loop counts are Bell-type inequalities, and they can be computed from field data. Each survey that sees one pair supplies one term, and a negative loop count says no single landscape fits the surveys. The gap to applicability is that each term needs paired observations over a comparable span, and real data are noisy, so a violation must be large to mean anything.

17 What the lived side asks of the mathematics

Questions 1–3 are the current priorities (October 2026). Question 4 waits on them.

1. **(priority)** *A finite, probabilistic version.* When are local marginals on a web of relations consistent yet unrealisable? Does composition, and not only cycles, govern it? *Partly answered:* the finite probabilistic version is contextuality, and the loop counts are its tests (Section 7). Composed pair by pair, answers determine a set of closing laws. Below the $3 \times 4 \times 4$ threshold the askable questions allow exactly the realisable ones; above it they allow more, and the excess is hidden incoherence.
2. **(priority)** *Scale as the source of "beyond".* In the theory, "beyond every number" creates twist. In ecology and society the analogue may be *separation of scales*. Can that be made precise? *Partly answered:* for probabilities, fineness alone suffices: hidden incoherence appears once two relations distinguish four values and the third three.
3. **(priority)** *Dynamics, revisited.* The theory found that motion only moves worlds. The lizard cycle is a *dynamic* order. Can a cycle in time, rather than in the questions, carry coherence without a top?
4. *Design.* Ostrom asked which institutions hold together. Could the theory say which webs of local rules can cohere without a centre, and which must fail by composition?

Questions to carry back into the world

1. Do my views form a loop?
2. Is there a joint question nobody can ask?
3. Is an idealisation hiding a contradiction?
4. When things fail to add up, is it a circle, or the way steps compose?
5. Am I asking about relations, or about relata?

References

[1] S. Abramsky and A. Brandenburger, The sheaf-theoretic structure of non-locality and contextuality, *New J. Phys.* 13 (2011) 113036.

[2] D. Aerts, J. Broekaert, M. Czachor, M. Kuna, B. Sinervo and S. Sozzo, Quantum structure in competing lizard communities, *Ecol. Model.* 281 (2014), 38–51.

[3] N. W. Barendregt and P. J. Thomas, Heteroclinic cycling and extinction in May–Leonard models with demographic stochasticity, *J. Math. Biol.* 86 (2023); arXiv:2111.05902.

[4] S. Allesina and J. M. Levine, A competitive network theory of species diversity, *PNAS* 108(14) (2011) 5638–5642.

[5] S. Abramsky and R. S. Barbosa, The logic of contextuality, *CSL 2021*, LIPIcs 183 (2021), 5:1–5:18; arXiv:2011.03064.

[6] M. Araújo, M. T. Quintino, C. Budroni, M. Terra Cunha and A. Cabello, All noncontextuality inequalities for the n-cycle scenario, *Phys. Rev. A* 88 (2013), 022118.

[7] Aristotle, *Physics*, Book III, ch. 6.

[8] C. Beeri, R. Fagin, D. Maier and M. Yannakakis, On the desirability of acyclic database schemes, *J. ACM* 30(3) (1983) 479–513.

[9] K. P. S. Bhaskara Rao and M. Bhaskara Rao, *Theory of Charges: A Study of Finitely Additive Measures*, Pure and Applied Mathematics 109, Academic Press, London, 1983.

[10] L. W. Buss and J. B. C. Jackson, Competitive networks: nontransitive competitive relationships in cryptic coral reef environments, *Amer. Natur.* 113 (1979) 223–234.

[11] A. Cabello, Simple explanation of the quantum violation of a fundamental inequality, *Phys. Rev. Lett.* 110 (2013), 060402.

[12] R. Choudhary and R. S. Barbosa, Exclusivity principle, Ramsey theory, and n-cycle PR boxes, arXiv:2411.09773 (2024).

[13] J. Dravécký and J. Šipoš, On the additivity of Gudder integral, *Math. Slovaca* 30(3) (1980) 299–303.

[14] P. Feyerabend, *Against Method*, New Left Books, London, 1975.

[15] T. Fritz, A. B. Sainz, R. Augusiak, J. B. Brask, R. Chaves, A. Leverrier and A. Acín, Local orthogonality as a multipartite principle for quantum correlations, *Nat. Commun.* 4 (2013), 2263.

[16] S. Gudder, Quantum probability spaces, *Proc. Amer. Math. Soc.* 21 (1969), 296–302.

[17] A. Horn and A. Tarski, Measures in Boolean algebras, *Trans. Amer. Math. Soc.* 64 (1948), 467–497.

[18] K. Gödel, Die Vollständigkeit der Axiome des logischen Funktionenkalküls, *Monatsh. Math. Phys.* 37 (1930) 349–360.

[19] G. Goldberg, On products of ultrafilters, *J. Symbolic Logic*, published online 2025, doi:10.1017/jsl.2025.10097; arXiv:2102.04677.

[20] L. H. Gunderson and C. S. Holling (eds.), *Panarchy: Understanding Transformations in Human and Natural Systems*, Island Press, Washington, DC, 2002.

[21] N. Hindman and D. Strauss, *Algebra in the Stone–Čech Compactification*, de Gruyter, 1998; 2nd ed. 2012.

[22] I. Kant, *Kritik der reinen Vernunft*, 1781/1787; Antinomy of Pure Reason, §§7–8.

[23] M. Katětov, A theorem on mappings, *Comment. Math. Univ. Carolinae* 8(3) (1967) 431–433.

[24] B. Kerr, M. A. Riley, M. W. Feldman and B. J. M. Bohannan, Local dispersal promotes biodiversity in a real-life game of rock–paper–scissors, *Nature* 418 (2002) 171–174.

[25] A. Kharoof, S. Ipek and C. Okay, Extremal simplicial distributions on cycle scenarios with arbitrary outcomes, arXiv:2406.19961.

[26] S. Kochen and E. P. Specker, The problem of hidden variables in quantum mechanics, *J. Math. Mech.* 17 (1967) 59–88.

[27] P. Kropotkin, *Mutual Aid: A Factor of Evolution*, William Heinemann, London, 1902.

[28] Larsen and Haine, Monte Carlo replication of competitive network theory, abstract, Ecological Society of America annual meeting, 2015 (conference abstract; not yet read in full).

[29] D. Maharam, Consistent extensions of linear functionals and of probability measures, *Proc. Sixth Berkeley Symp. Math. Statist. Probab.*, Vol. II (1972), 127–147.

[30] E. Minguzzi, Spacelike slices from globally well-behaved simultaneity connections, arXiv:gr-qc/0501055 (2005).

[31] R. M. May and W. J. Leonard, Nonlinear aspects of competition between three species, *SIAM J. Appl. Math.* 29(2) (1975) 243–253.

[32] A. Gaunersdorfer, Time averages for heteroclinic attractors, *SIAM J. Appl. Math.* 52 (1992) 1476–1489.

[33] F. Takens, Heteroclinic attractors: time averages and moduli of topological conjugacy, *Bol. Soc. Bras. Mat.* 25 (1994) 107–120.

[34] M. Navara, Two-valued states on a concrete logic and the additivity problem, *Math. Slovaca* 34(3) (1984) 329–336.

[35] M. Navara and P. Pták, Two-valued measures on σ-classes, *Časopis Pěst. Mat.* 108(3) (1983) 225–229.

[36] E. Ostrom, *Governing the Commons: The Evolution of Institutions for Collective Action*, Cambridge University Press, 1990.

[37] E. Ostrom, Beyond markets and states: polycentric governance of complex economic systems, *Amer. Econ. Rev.* 100(3) (2010) 641–672.

[38] R. Penrose, On the cohomology of impossible figures, *Structural Topology* 17 (1991) 11–16; reprinted in *Leonardo* 25 (1992).

[39] G. Rizzi and M. L. Ruggiero, A direct kinematical derivation of the relativistic Sagnac effect for light or matter beams, arXiv:gr-qc/0306128 (2003).

[40] S. Sullivant, Compressed polytopes and statistical disclosure limitation, *Tohoku Math. J.* 58 (2006), 433–445.

[41] H. Ohsugi and T. Hibi, Non-very ample configurations arising from contingency tables, *Ann. Inst. Statist. Math.* 62 (2010), 639–644.

[42] B. Sinervo and C. M. Lively, The rock–paper–scissors game and the evolution of alternative male strategies, *Nature* 380 (1996) 240–243.

[43] E. Specker, Die Logik nicht gleichzeitig entscheidbarer Aussagen, *Dialectica* 14 (1960) 239–246.

[44] A. Atserias and P. G. Kolaitis, Consistency, acyclicity, and positive semirings, arXiv:2009.09488 (2020), Lemmas 13–15.

[45] N. N. Vorob'ev, Consistent families of measures and their extensions, *Theory Probab. Appl.* 7(2) (1962) 147–163.

[46] M. J. Wainwright and M. I. Jordan, Graphical models, exponential families, and variational inference, *Found. Trends Mach. Learn.* 1(1–2) (2008) 1–305.

Full proofs and status: `pointless_coherence.tex`. The theory's current shape: `theory_shape.md`. The why: `WHY.md`.